SOFTWARE ENGINEERING LAB – 3

Architectural Style Justification

Problem Statement 001 – Smart Lab Equipment Slot Reservation Portal | Team 001 | UE24CS341A

Scope: Epic 2 – Calibration & Technician Management (FR-002, FR-005) from Lab 2

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Architecture Selection

We chose **Layered Architecture** for the Smart Lab Equipment Slot Reservation Portal.

Architectural Choice

Layered Architecture (Presentation → Business → Data) was selected after comparing it with Microservices and Client-Server. Microservices would split calibration and equipment data across services and add network and operational overhead, and plain Client-Server would mix the calibration and technician rules with the user interface. Layered Architecture keeps the Web Portal UI, the business rules (Reservation Manager, Calibration Service, Equipment Management) and the database separate, which suits a single-lab system with moderate load.

Reason 1 – Calibration data must stay consistent between booking and technician updates (FR-002, FR-005)

A reservation may be confirmed only when the equipment has valid calibration, and lab technicians change that calibration status and expiry. In a layered design both sides use one business layer and one central database. Equipment Management writes the status through IEquipmentData, and the Calibration Service reads the same record through ICalibrationData whenever the Reservation Manager calls ICalibrationCheck. A technician's update is therefore seen by the very next booking check, and equipment with expired calibration cannot be confirmed. With Microservices the calibration data would have to be synchronised between services.

Reason 2 – Clear separation that maps to the requirements

Each requirement has one home: the calibration check and the block on invalid or expired equipment (FR-002) sit in the Calibration Service and the Reservation Manager, and the technician updates (FR-005) sit in Equipment Management. A change to the calibration rule touches only the Calibration Service. The known drawbacks of layering, such as extra overhead and coupling between layers, are small at this scale, while the independent scaling of Microservices would add complexity without real benefit for one lab.

Security Advantage

Only the business layer can reach the database, so the browser never touches stored data directly. The external Authentication Service issues a token and role when a user logs in through IAuth, and Equipment Management validates that token and the technician role on the server for every update request. As a result, only authorized lab technicians can change equipment availability or calibration status, and students and other users cannot tamper with equipment records.

Performance Benefit

A booking follows a short, direct path: Web Portal UI → Reservation Manager → one internal call to the Calibration Service → one database lookup of status and expiry. There are no extra network hops between many separate services. Because the portal and the technicians read and write the same database, a status change is visible to students immediately without any cache or data synchronisation delay.